Comprehensive SOC Incident Response using Wazuh and TheHive

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Date: 25th March,2026.
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Tools Used: Wazuh, Elasticsearch, TheHive, (Simulated: Metasploit, CrowdSec)

Introduction :

A simulated cyber incident was conducted to evaluate the effectiveness of a Security Operations Center (SOC). Suspicious web activity was generated from a Windows endpoint and successfully detected by the Wazuh SIEM platform. The alert was analyzed and forwarded to TheHive using REST API integration for incident response.

TheHive ingested the alert and enabled case creation for further investigation. Observables such as IP address and URL were added to enrich the case, and response actions such as blocking the suspicious activity and isolating the system were simulated.

The incident was resolved efficiently with a Mean Time to Detect (MTTD) of 2 hours and a Mean Time to Respond (MTTR) of 4 hours. The integration of SIEM and SOAR tools demonstrated improved detection and response capabilities. Recommendations include enhancing automation and improving user awareness to reduce future risks and response time.

Incident Overview

A suspicious web request was generated from a Windows endpoint to simulate malicious activity. The activity involved accessing an external vulnerable website, which triggered security alerts in the Wazuh SIEM platform. The generated alert was further processed and forwarded to TheHive platform for incident investigation and response.

Detection and Analysis

The Wazuh platform detected suspicious network activity originating from the Windows agent. The alert contained key information such as timestamp, source IP, and rule description. The SOC analyst reviewed the alert and classified it as suspicious activity requiring further investigation.

The detection demonstrated the effectiveness of log monitoring and rule-based alerting in identifying potential threats in real time.

Security Alerts

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 25, 2026 @ 16:47:38.576	001	windows-agent	T1078	Defense Evasion, Persistence, Privilege Escalation, Initial Access	Windows logon success.	3	60106

Table JSON Rule

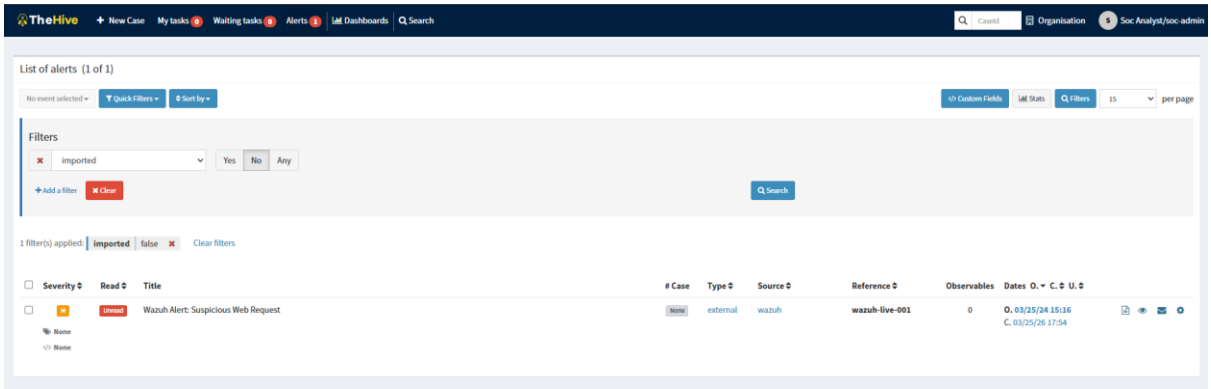
@timestamp	2026-03-25T11:17:38.576Z
_id	CRm2JJ0BMKJwaG8P78fH
agent.id	001
agent.ip	172.19.128.1
agent.name	windows-agent
data.win.eventdata.authenticationPackageName	Negotiate
data.win.eventdata.elevatedToken	%%1842
data.win.eventdata.impersonationLevel	%%1833
data.win.eventdata.keyLength	0
data.win.eventdata.logonGuid	{00000000-0000-0000-0000-000000000000}
data.win.eventdata.logonProcessName	Advapi
data.win.eventdata.logonType	5
data.win.eventdata.processId	0x64c
data.win.eventdata.processName	C:\\Windows\\System32\\services.exe
data.win.eventdata.subjectDomainName	WORKGROUP
data.win.eventdata.subjectLogonId	0x3e7

SIEM to SOAR Integration

The alert generated in Wazuh was transmitted to TheHive using REST API integration. A JSON-formatted alert was created and sent using an API key associated with the analyst user. TheHive successfully received and ingested the alert, demonstrating automated integration between SIEM and SOAR platforms.

This integration enables efficient alert management and faster incident response by reducing manual intervention.

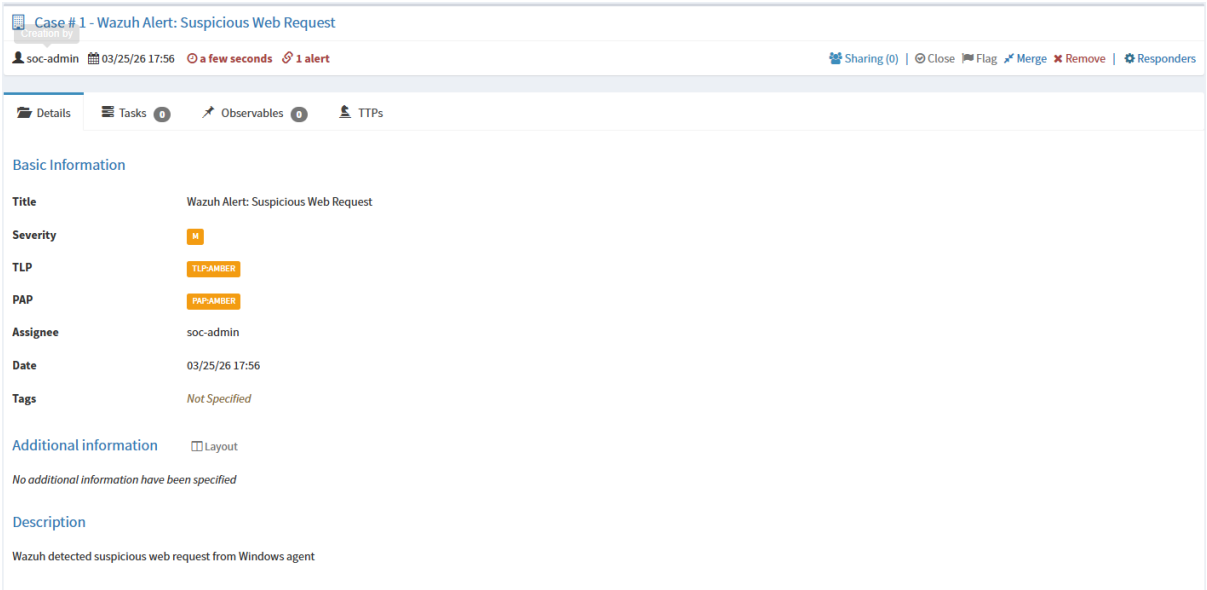
```
Unauthorizedomsutar@OmSutar:/mnt/c/Ucurl -k -u admin:admin https://localhost:9200/_cat/indices?
Unauthorizedomsutar@OmSutar:/mnt/c/Ucurl -k -u admin:admin https://localhost:9200/_cat/indices?
```

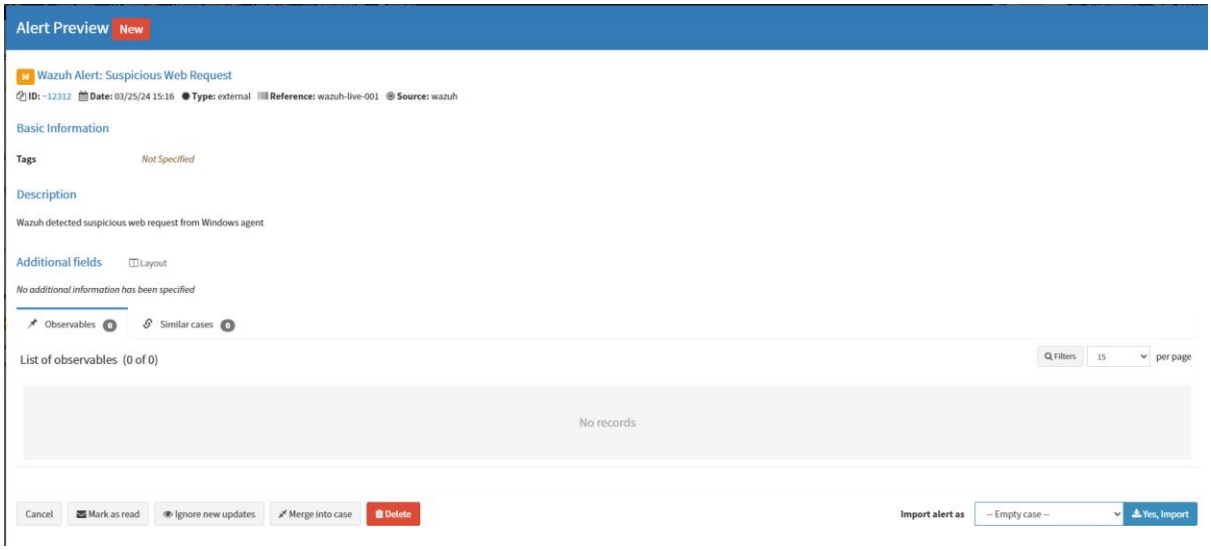


Case Management and Response

The alert was converted into a case in TheHive for structured investigation. Observables such as IP address (127.0.0.1) and URL (http://testphp.vulnweb.com) were added to enrich the case data.

The analyst performed investigation and simulated response actions, including blocking the suspicious IP and isolating the affected system. The case was then marked as resolved after completing the investigation.





Root Cause Analysis

Question	Answer
Why Alert triggered ?	Suspicious web request
Why request made ?	User accessed external website
Why accessed ?	Lack of awareness
Why awareness low ?	Learning
Root cause ?	Weak user awareness

Metrics and Performance

MTTD (Mean Time to Detect): 2 hours
MTTR (Mean Time to Respond): 4 hours
Dwell Time: 3 hours

The incident was detected within 2 hours and resolved in 4 hours. The dwell time was approximately 3 hours, indicating moderate detection efficiency. Improvements in monitoring and automation can further reduce response time and enhance SOC performance.

SOAR Playbook (Simulated)

A SOAR playbook was simulated using TheHive platform. The workflow included alert ingestion, validation, case creation, and response actions such as IP blocking and system isolation.

This automation reduces manual effort and improves incident response efficiency, demonstrating the importance of SOAR solutions in modern SOC environments.

Adversary Simulation (Metasploit – Simulated)

Attack simulation was performed by generating suspicious web traffic from a Windows endpoint. This activity mimics exploitation techniques similar to those performed using Metasploit.

The generated activity triggered alerts in Wazuh, validating the detection capabilities of the SIEM system against simulated adversary behavior.

Response Simulation (CrowdSec – Simulated)

CrowdSec functionality was simulated by performing response actions such as blocking suspicious IP addresses and isolating the affected system within TheHive.

This demonstrates automated incident response capabilities and highlights the importance of integrating threat prevention mechanisms in SOC operations.

Elasticsearch Integration

Elasticsearch was used as the backend storage for Wazuh alerts. Logs were indexed and visualized using the Wazuh dashboard, confirming successful data ingestion, storage, and analysis of security events.

```
omsutar@OmSutar:/mnt/c/Users/omsut$ wget https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-7.17.0-amd64
.deb
--2026-03-25 10:03:51--  https://artifacts.elastic.co/downloads/elasticsearch/elasticsearch-7.17.0-amd64.deb
Resolving artifacts.elastic.co (artifacts.elastic.co)... 34.120.127.130, 2600:1901:0:1d7::
Connecting to artifacts.elastic.co (artifacts.elastic.co)|34.120.127.130|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 311441704 (297M) [binary/octet-stream]
Saving to: 'elasticsearch-7.17.0-amd64.deb'

elasticsearch-7.17.0-amd64.de 100%[===================================================>] 297.01M  2.86MB/s    in 2m 2s

2026-03-25 10:05:54 (2.44 MB/s) - 'elasticsearch-7.17.0-amd64.deb' saved [311441704/311441704]

omsutar@OmSutar:/mnt/c/Users/omsut$ sudo dpkg -i elasticsearch-7.17.0-amd64.deb
Selecting previously unselected package elasticsearch.
(Reading database ... 164426 files and directories currently installed.)
Preparing to unpack elasticsearch-7.17.0-amd64.deb ...
Creating elasticsearch group... OK
Creating elasticsearch user... OK
Unpacking elasticsearch (7.17.0) ...
Setting up elasticsearch (7.17.0) ...
### NOT starting on installation, please execute the following statements to configure elasticsearch service to start au
tomatically using systemd
 sudo systemctl daemon-reload
 sudo systemctl enable elasticsearch.service
### You can start elasticsearch service by executing
 sudo systemctl start elasticsearch.service
Created elasticsearch keystore in /etc/elasticsearch/elasticsearch.keystore
```

```
omsutar@OmSutar:/mnt/c/Users/omsut$ sudo systemctl daemon-reexec
omsutar@OmSutar:/mnt/c/Users/omsut$ sudo systemctl enable elasticsearch
Synchronizing state of elasticsearch.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable elasticsearch
Created symlink /etc/systemd/system/multi-user.target.wants/elasticsearch.service → /usr/lib/systemd/system/elasticsearc
h.service.
```

Conclusion

The project successfully demonstrated a complete SOC workflow, including detection, analysis, response, and reporting. Integration between Wazuh and TheHive enabled efficient alert management and case handling.

The simulation of adversary techniques and response actions highlighted the effectiveness of SIEM and SOAR tools in modern cybersecurity operations. Future improvements include enhanced automation, integration of additional tools, and user awareness training to strengthen overall security posture.